Algorithms 2

Linear Programming

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1. Linear Programming

1.1 What is Linear Programming?

- Linear programming is an optimization model with a linear objective and linear constraints.

$$b_1 \leq x_1 a_{1,1} + x_2 a_{2,1} + \ldots + x_n a_{n,1}$$

$$b_2 \leq x_1 a_{1,2} + x_2 a_{2,2} + \ldots + x_n a_{n,2}$$

$$\ldots$$

$$b_m \underbrace{\leq}_{\substack{\text{can also} \\ \text{be } = \text{ or } \geq}} x_1 a_{1,m} + x_2 a_{2,m} + \ldots + x_n a_{n,m}$$

If we see $=$ it must be $=$ for all, if we see $\leq$ or $\geq$ likewise.

- Variables may be continuous, and every constraint is either an equality or inequality.

$$\underbrace{x_1, \ldots x_n \in \mathbb{Z}}_{\text{Integer form}} \text{ or } \underbrace{x_1, \ldots x_n \in \mathbb{R}}_{\text{Linear form}}$$

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&\ldots \\
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Example: b_j is the current electricity demand of the j-th node; $a_{i,j} x_i$ is the amount of electricity from plant i used at node j; and x_i is the total electricity generated by plant i.

Goal: Calculate the values of $x_1, \ldots, x_n$ such that all constraints are satisfied. i.e. we meet the electricity demand. Additionally, we would also prefer $\sum_{i=1}^{n} x_i c_i$ to be minimized. **(c_i might be the cost of using the i-th power plant.)**

Definition 1.2.1 An Integer program is an optimization problem over $\boldsymbol{x} \in \mathbb{Z}^n$ that can be written in one of the following six forms:

$$\max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} \leq \boldsymbol{b}\} \qquad \max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} \leq \boldsymbol{b}, \boldsymbol{x} \geq 0\} \qquad \max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq 0\}$$
$$\min\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} \geq \boldsymbol{b}\} \qquad \min\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} \geq \boldsymbol{b}, \boldsymbol{x} \geq 0\} \qquad \min\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq 0\}$$

Definition 1.2.2 A linear program is an optimization problem over $\boldsymbol{x} \in \mathbb{R}^n$ that can be written in one of the following six forms:

$$\max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} \leq \boldsymbol{b}\} \qquad \max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} \leq \boldsymbol{b}, \boldsymbol{x} \geq 0\} \qquad \max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq 0\}$$
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Remark

LP's $\in$ P but IP's $\in$ NPH

2. Examples

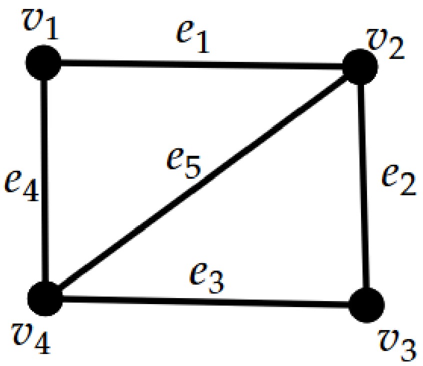
2.1 Incidence matrix

The *incidence* matrix of a graph G is given by

$$\begin{array}{c} \\ v_1 \\ v_2 \\ \vdots \\ v_n \end{array}\begin{array}{c} e_1\ e_2\ \dots\ e_m \\ \left[\begin{array}{ccc} & & \\ & 0/1 & \\ & & \\ & & \end{array}\right] \end{array} \qquad M_{ve} := \begin{cases} 1, & v \in e \\ 0, & v \notin e \end{cases}$$

$$M := M(G)$$

$$\begin{array}{c|c} & \begin{array}{ccccc} e_1 & e_2 & e_3 & e_4 & e_5 \end{array} \\ \begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} & \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{array}$$



Each column has #1 = 2
Each row has #1 = $\deg(v)$

Given G with n vertices and m edges, we seek a vector $\boldsymbol{x} \in \{0, 1\}^m$ s.t.

- $x_e = 1$ when edge e belongs to the matching, and $x_e = 0$ otherwise.
- maximize $\sum_{e \in E} x_e = \max \mathbf{1}^t \boldsymbol{x}$. **(We want maximum matching)**

This vector $\boldsymbol{x}$ is the characteristic vector of a matching in G.

So far the above doesn't define a matching!
We need to define constraints.

2.2 Maximum Matching

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or by matricial form

$$\boxed{\begin{array}{c} \boldsymbol{x} \in \mathbb{Z}^n \\ \max \mathbf{1}^t \boldsymbol{x} \end{array}}$$

$$\begin{array}{c} \\ v_1 \\ v_2 \\ \vdots \\ v_n \end{array} \overset{\begin{array}{cccccc} e_1 & e_2 & & \cdots & & e_m \end{array}}{\underset{\substack{M \\ n \times m}}{\begin{bmatrix} 1 & 1 & 0 & \cdots & 0 & 1 \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix}}} \underset{\substack{x \\ m \times 1}}{\begin{bmatrix} 1 \\ 0 \\ 1 \\ \vdots \\ 1 \\ 1 \end{bmatrix}} = \underset{\substack{Mx \\ n \times 1}}{\begin{bmatrix} (Mx)_1 \\ \vdots \\ (Mx)_i \\ \vdots \end{bmatrix}}$$

$(Mx)_i = \langle v_i, x \rangle$

#edges incident to v_i by the given matching

This is why $Mx \leq 1$ must hold if X is a matching!

2.2 Maximum Matching

For the max-matching problem, we arrive at the following

IP for maximum matching

$$\max_{x \in \mathbb{Z}^m} \{\mathbf{1}^t \boldsymbol{x} : M\boldsymbol{x} \leq \mathbf{1}, \boldsymbol{x} \geq \mathbf{0}\}$$

LP for maximum matching

$$\max_{\boldsymbol{x} \in \mathbb{R}^m} \{\mathbf{1}^t \boldsymbol{x} : M\boldsymbol{x} \leq \mathbf{1}, \boldsymbol{x} \geq \mathbf{0}\}$$

This called the:
Corresponding LP Relaxation

Remark

We dont need to bound $\boldsymbol{x} \leq \mathbf{1}$ as if there exists i for which $x_i > 1$. Then, there must exist $j \in [m]$ for which $M_{j,i} < 1$. But, we know that M is the *incidence matrix* of G so that for exactly two indicies j_1, j_2 we have $M_{j_1,i} = 1$ and $M_{j_1,i} = 1$.

2.3 Vertex Cover

Given G with n vertices and m edges, we seek a vector $\boldsymbol{y} \in \{0,1\}^n$ s.t.

- $y_v = 1$ when vertex v belongs to the **vc**, and $y_v = 0$ otherwise.
- minimize $\sum_{v \in V} y_v = \min \mathbf{1}^t \boldsymbol{y}$. **(We want minimum vertex-cover)**

Constaints: $\forall e \in E : \sum_{v \in e} y_v \geq 1$ **(Every edge is covered at least once)**

This vector $\boldsymbol{y}$ is the characteristic vector of a vertex cover in G.

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or by matricial form

$$\boxed{\begin{array}{c} \boldsymbol{y} \in \mathbb{Z}^n \\ \min \mathbf{1}^t \boldsymbol{y} \end{array}}$$

$$\underset{\substack{M^t \\ n \times m}}{\begin{array}{cc} & \begin{array}{cccccc} v_1 & v_2 & & \cdots & & v_n \end{array} \\ \begin{array}{c} e_1 \\ e_2 \\ \vdots \\ e_m \end{array} & \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 1 \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}} \underset{\substack{y \\ n \times 1}}{\begin{bmatrix} 1 \\ 1 \\ 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix}} = \underset{\substack{My \\ m \times 1}}{\begin{bmatrix} (M^t y)_1 \\ \vdots \\ (M^t y)_i \\ \vdots \end{bmatrix}}$$

$$\boxed{\begin{array}{c} (M^t y)_i = \langle e_i, y \rangle \\ \text{\#vertices covering the edge } e_i \end{array}}$$

This is why $M^t y \geq 1$ must hold if Y is a matching!

2.3 Vertex Cover

For the min-vc problem, we arrive at the following

IP for minimum-vc

$$\min_{\boldsymbol{y}\in\mathbb{Z}^n}\{\mathbf{1}^t\boldsymbol{y} : M^t\boldsymbol{y} \geq \mathbf{1}, \boldsymbol{y} \geq \mathbf{0}\}$$

LP for minimum-vc

$$\underbrace{\min_{\boldsymbol{y}\in\mathbb{R}^n}\{\mathbf{1}^t\boldsymbol{y} : M\boldsymbol{y} \geq 1, \boldsymbol{y} \geq 0\}}_{\substack{\text{This called the:} \\ \textbf{Corresponding LP Relaxation}}}$$

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What if we wanted vc with minimum weight? i.e. every vertex $v \in V$ has weight $w_v \in \mathbb{R}$?

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$$\min_{\boldsymbol{y}\in\mathbb{Z}^n}\left\{\underbrace{\boldsymbol{w}^t\boldsymbol{y}}_{\sum_{v\in V} w_v \cdot y_v} : M^t\boldsymbol{y} \geq \mathbf{1}, \boldsymbol{y} \geq \mathbf{0}\right\}$$

2.4 Vertex Cover vs Maximum Matching

By Konig we know that in bipartite graphs it must hold

IP for maximum matching

$$\max_{\boldsymbol{x}\in\mathbb{Z}^n}\{\mathbf{1}^t\boldsymbol{x} : M\boldsymbol{x} \leq \mathbf{1}, \boldsymbol{x} \geq \mathbf{0}\}$$

$$=$$

IP for minimum-vc

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If G is not bipartite, then in fact

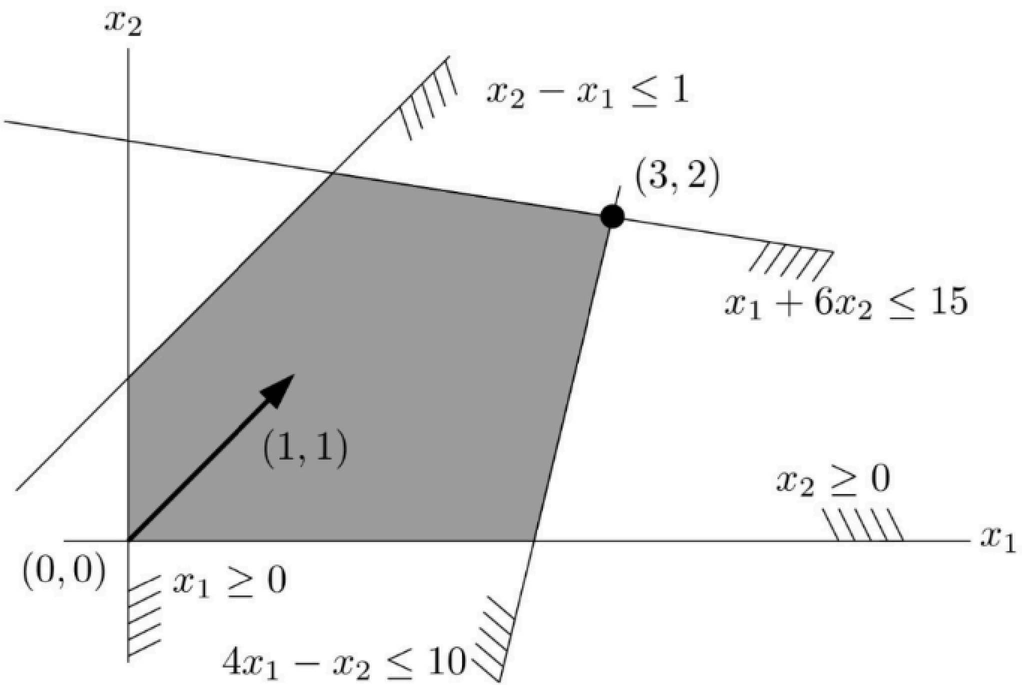
IP for maximum matching		**IP** for minimum-vc	
$\max_{\boldsymbol{x}\in\mathbb{Z}^n}\{\mathbf{1}^t\boldsymbol{x} : M\boldsymbol{x} \le \mathbf{1}, \boldsymbol{x} \ge \mathbf{0}\}$	$\in \mathrm{P}$	$\min_{\boldsymbol{y}\in\mathbb{Z}^n}\{\mathbf{1}^t\boldsymbol{y} : M^t\boldsymbol{y} \ge \mathbf{1}, \boldsymbol{y} \ge \mathbf{0}\}$	$\in \mathrm{NPC}$

3. Geometric analysis of LP’s

3.1 Geometric analysis of LP's

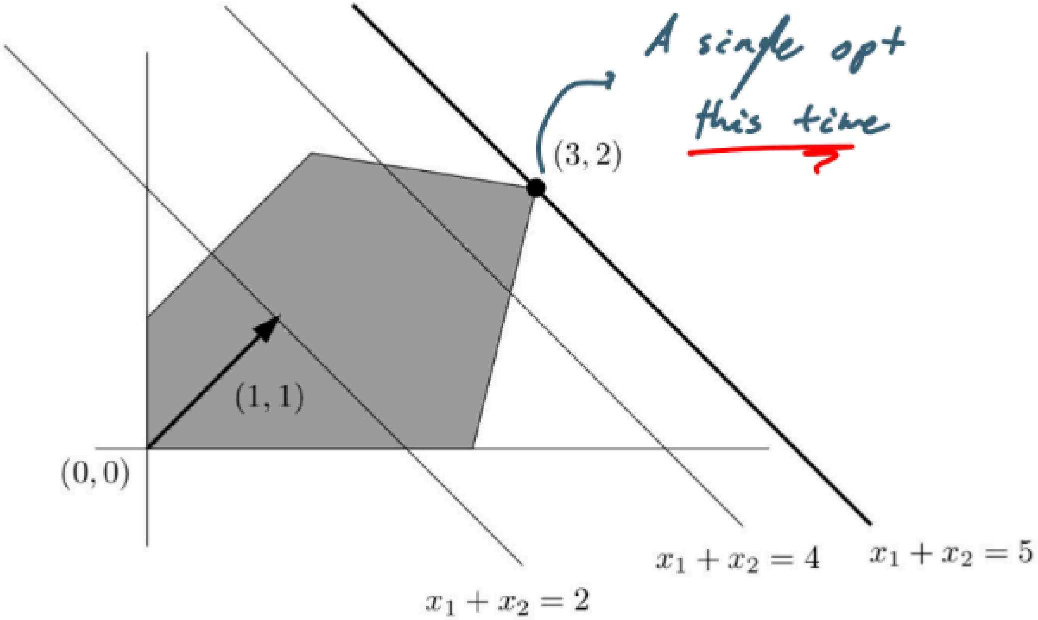
Consider the following **LP**:

$$\begin{aligned} &\max x_1 + x_2 \\ &s.b.t. \\ &\qquad -x_1 + x_2 \leq 1 \\ &\qquad x_1 + 6x_2 \leq 15 \\ &\qquad 4x_1 - x_2 \leq 10 \\ &\qquad x_1 \geq 0 \\ &\qquad x_2 \geq 0 \end{aligned}$$



$$\max_{x \in \mathbb{R}^2} (1,1) \cdot (x_1, x_2) \text{ s.t. } \begin{pmatrix} -1 & 1 \\ 4 & 6 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 1 \\ 15 \\ 10 \end{pmatrix}, \quad (x_1, x_2) \geq (0, 0)$$

Solving the problem Geometrically, would mean taking the line $x_1 + x_2 = c$ and moving it in the direction $(1,1)$ until we get out of constraint.



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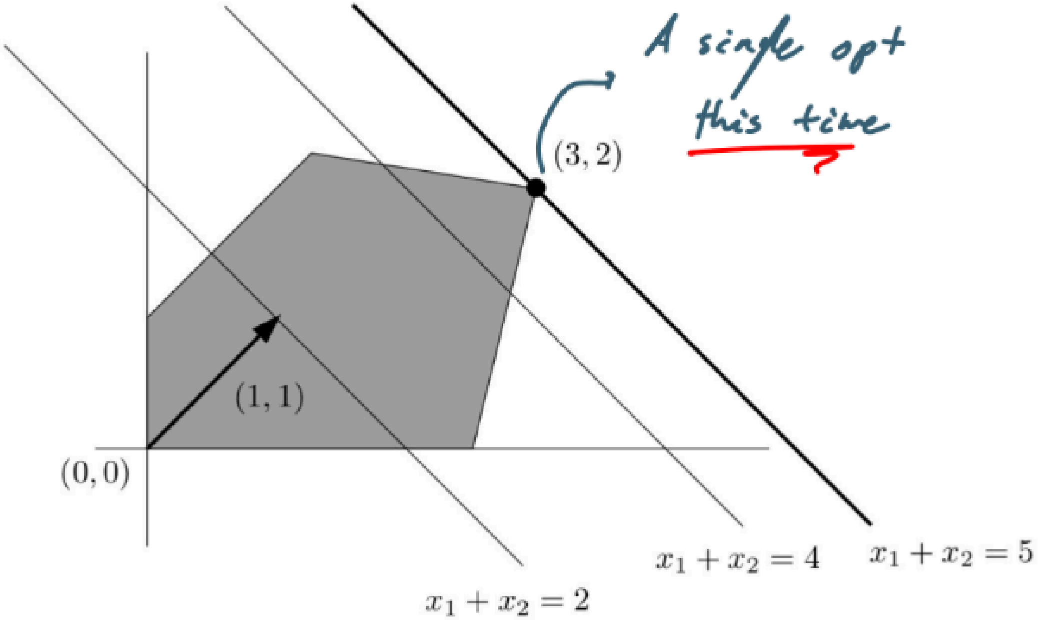

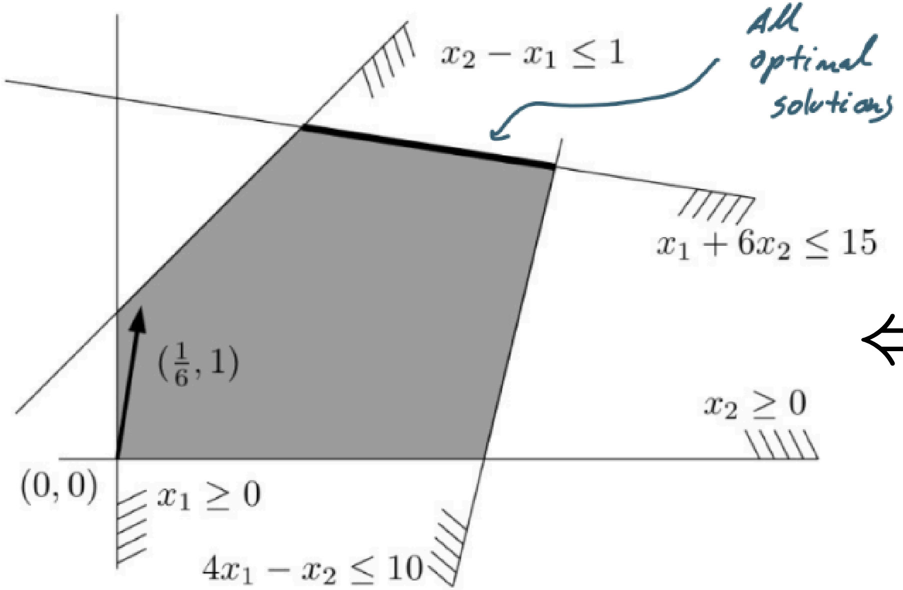

$\Longleftarrow$ If we change the constraint to max $\frac{1}{6}x_1 + x_2$.
Then, we have infinitely maxny solutions.

3.2 Standard forms of LP’s

Definition 3.2.1 A Linear program in a one of the following 2-forms:

$$\max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq \boldsymbol{0}\} \qquad \min\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq \boldsymbol{0}\}$$

is said to be in **equational form**;

Remark

Optimal solution for **LP’s** in **equational form** are the easiest to exaplain and we will focus on those.